

1) If $Z = \frac{xy}{x^2+y^2}$, find $\frac{\partial Z}{\partial x}$ and $\frac{\partial Z}{\partial y}$

Ans

$$\frac{\partial Z}{\partial x} = \frac{(x^2+y^2) \cdot y - (xy) \cdot 2x}{(x^2+y^2)^2}$$

$$\frac{\partial Z}{\partial y} = \frac{(x^2+y^2) \cdot x - (xy) \cdot 2y}{(x^2+y^2)^2}$$

2

Show that the equation $u(x,t) = \sin(x-ct)$ satisfies wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Ans.

$$\frac{\partial u}{\partial t} = \cos(x-ct) \cdot -c$$

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= -\sin(x-ct) \cdot (-c) \cdot (-c) \\ &= -c^2 \sin(x-ct) \end{aligned}$$

$$\frac{\partial u}{\partial x} = \cos(x-ct)$$

$$\frac{\partial^2 u}{\partial x^2} = -\sin(x-ct)$$

$$\therefore \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$\begin{aligned} &= -c^2 \sin(x-ct) = c^2 \cdot -\sin(x-ct) \\ &= -c^2 \sin(x-ct) \end{aligned}$$

3a) If $W = F(P, Q, R)$ where $P = x - y$, $Q = y - z$, $R = z - x$ Prove that

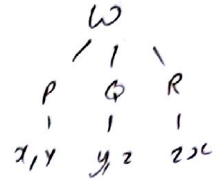
$$\frac{\partial W}{\partial x} + \frac{\partial W}{\partial y} + \frac{\partial W}{\partial z} = 0$$

Ans a)

Let $P = x - y$

$Q = y - z$

$R = z - x$



$$\frac{\partial W}{\partial x} = \frac{\partial W}{\partial P} \cdot \frac{\partial P}{\partial x} + \frac{\partial W}{\partial R} \cdot \frac{\partial R}{\partial x}$$

$$= \frac{\partial W}{\partial P} \cdot 1 + \frac{\partial W}{\partial R} \cdot -1$$

$$= \frac{\partial W}{\partial P} - \frac{\partial W}{\partial R}$$

$$\frac{\partial W}{\partial y} = \frac{\partial W}{\partial P} \cdot \frac{\partial P}{\partial y} + \frac{\partial W}{\partial Q} \cdot \frac{\partial Q}{\partial y}$$

$$= \frac{\partial W}{\partial P} \cdot -1 + \frac{\partial W}{\partial Q} \cdot 1$$

$$= -\frac{\partial W}{\partial P} + \frac{\partial W}{\partial Q}$$

$$\frac{\partial W}{\partial z} = \frac{\partial W}{\partial Q} \cdot \frac{\partial Q}{\partial z} + \frac{\partial W}{\partial R} \cdot \frac{\partial R}{\partial z}$$

$$= \frac{\partial W}{\partial Q} \cdot -1 + \frac{\partial W}{\partial R} \cdot 1$$

$$= -\frac{\partial W}{\partial Q} + \frac{\partial W}{\partial R}$$

$$\frac{\partial W}{\partial x} + \frac{\partial W}{\partial y} + \frac{\partial W}{\partial z} = 0$$

$$\therefore \frac{\partial W}{\partial P} - \frac{\partial W}{\partial R} + -\frac{\partial W}{\partial P} + \frac{\partial W}{\partial Q} + -\frac{\partial W}{\partial Q} + \frac{\partial W}{\partial R} = \underline{\underline{0}}$$

3) Locate all relative extrema and saddle points of $f(x,y) = 4xy - x^4 - y^2$

Ans) $F(x,y) = 4xy - x^4 - y^2$

$$F_x = 4y - 4x^3 = 0 \quad (1)$$

$$F_y = 4x - 4y^3 = 0 \quad (2)$$

$$4y - 4x^3 = 0$$

$$4y = 4x^3$$

$$y = x^3$$

Substitute $y = x^3$ in (2)

$$4x - 4y^3 = 0$$

$$4x - 4(x^3)^3 = 0$$

$$4x - 4x^9 = 0$$

$$x - x^9 = 0$$

$$x(1 - x^8) = 0$$

$$x = 0, x^8 = 1$$

$$x = 0, x = \pm 1$$

When $x = 0$

$$4y - 4x^3 = 0$$

$$4y - 4 \cdot 0 = 0$$

$$4y = 0$$

$$y = 0$$

When $x = 1$

$$4y - 4x^3 = 0$$

$$4y - 4 \cdot 1^3 = 0$$

$$4y = 4$$

$$y = 1$$

When $x = -1$

$$4y - 4x(-1)^3 = 0$$

$$4y + 4 = 0$$

$$4y = -4$$

$$y = -1$$

$$f_{xx} = -4 \cdot 3x^2 = -12x^2$$

$$f_{yy} = -12y^2$$

$$f_{xy} = 4$$

$(0,0), (1,1), (-1,-1)$ are critical points

Critical points	f_{xx}	f_{yy}	f_{xy}	$D = f_{xx}f_{yy} - (f_{xy})^2$
$(0,0)$	0	0	4	-16
$(1,1)$	-12	-12	4	128
$(-1,-1)$	12	12	4	128

At $(0,0)$ $D = -16 < 0$ So function has Saddle point

At $(1,1)$ $D = 128 > 0, f_{xx} < 0$ function has relative maximum

At $(-1,-1)$ $D = 128 > 0, f_{xx} > 0$ function has relative minimum